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2.5.2 The Vector Product of Two Vectors

Definition

The vector product or cross product of two vectors is defined as another vector having a magnitude equal to the product of the magnitudes of two vectors and the sine of the angle between them. The direction of the product vector is perpendicular to the plane containing the two vectors, in accordance with the right hand screw rule or right hand thumb rule (Figure 2.22).

Thus, if $\vec{A}$ and $\vec{B}$ are two vectors, then their vector product is written as $\vec{A}\times\vec{B}$ which is a vector $\vec{C}$ defined by

$$\vec{C} = \vec{A}\times\vec{B} = (AB\sin\theta)\hat{n}$$

The direction $\hat{n}$ of $\vec{A}\times\vec{B}$, i.e., $\vec{C}$ is perpendicular to the plane containing the vectors $\vec{A}$ and $\vec{B}$ and is in the sense of advancement of a right handed screw rotated from $\vec{A}$ (first vector) to $\vec{B}$ (second vector) through the smaller angle between them. Thus, if a right-handed screw whose axis is perpendicular to the plane formed by $\vec{A}$ and $\vec{B}$, is rotated from $\vec{A}$ to $\vec{B}$ through the smaller angle between them, then the direction of advancement of the screw gives the direction of $\vec{A}\times\vec{B}$ i.e. $\vec{C}$ which is illustrated in Figure 2.22.

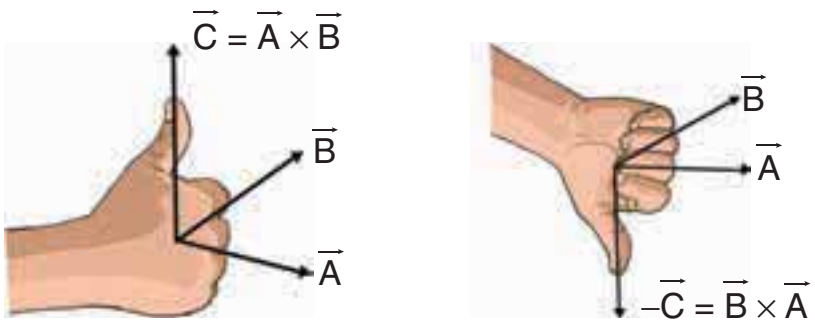
Note

According to Right Hand Rule, if the curvature of the fingers of the right hand represents the sense of rotation of the object, then the thumb, held perpendicular to the curvature of the fingers, represents the direction of the resultant $\vec{C}$.

VECTOR PRODUCT ("CROSS" PRODUCT)

The vector product of $\vec{A}$ and $\vec{B}$, written as $\vec{A}\times\vec{B}$, produces a third vector $\vec{C}$ whose magnitude is

$$\vec{C} = \vec{A}\times\vec{B} = |\vec{A}||\vec{B}|\sin\theta\,\hat{n} \qquad -\vec{C} = \vec{B}\times\vec{A}$$



$$\vec{A}\times\vec{B} = -(\vec{B}\times\vec{A})$$

Figure 2.22 Vector product of two vectors